



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2016 HSC Course Assessment Task 4 (Trial Examination)

Friday August 12, 2016

General instructions

- Working time – 2 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 11)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

BOSTES NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M1 – Mr Sekaran

☐ 12M3 – Mr Lam

☐ 12M4 – Mr Wall

☐ 12M2 – Mrs Bhamra

☐ 12M5 – Mrs Gan

Marker's use only.

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\frac{\quad}{10}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{70}$	

Section I

10 marks

Attempt Question 1 to 10

Mark your answers on the answer grid provided (labelled as page 11).

Questions

Marks

1. What is the number of asymptotes on the graph of $y = \frac{1}{x^2 - 1}$? 1

(A) 1 (B) 2 (C) 3 (D) 4

2. A is the point $(-1, 4)$ and B is $(9, -6)$. Which of the following are coordinates of a point that divides AB internally in the ratio $3 : 2$? 1

(A) $(1, 3)$ (B) $(3, 1)$ (C) $(-2, 5)$ (D) $(5, -2)$

3. The equation of the normal to the parabola $x^2 = 4ay$ at the variable point $P(2ap, ap^2)$ is given by $x + py = 2ap + ap^3$. 1

How many different values of p are there such that the normal passes through the focus of the parabola?

(A) 0 (B) 1 (C) 2 (D) 3

4. A curve has parametric equations $x = t - 3$ and $y = t^2 + 2$. 1

What is the Cartesian equation of this curve?

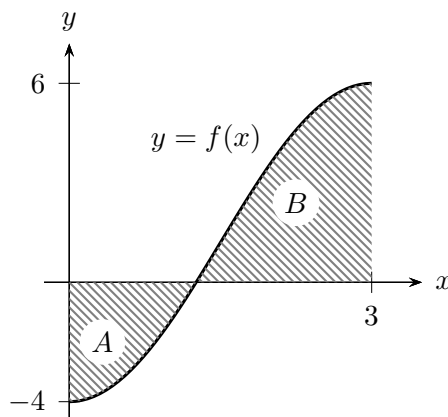
(A) $y = x^2 - x - 1$ (C) $y = x^2 - 6x + 11$

(B) $y = x^2 + x - 1$ (D) $y = x^2 + 6x + 11$

5. What is the value of $\tan^{-1}\left(\tan \frac{5\pi}{4}\right)$? 1

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $-\frac{5\pi}{4}$

6. A graph of the function $y = f(x)$ is shown. 1



Area A is equal to 3 square units, and area B is equal to 7 square units.

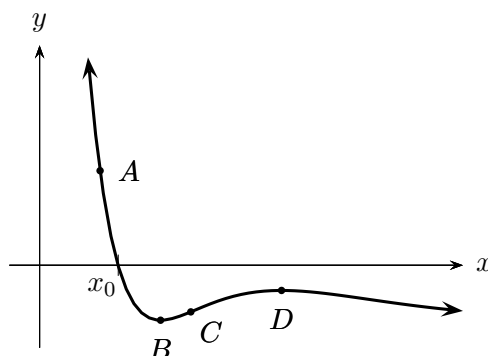
What is the value of $\int_{-4}^6 f^{-1}(x) dx$?

(A) 4 (B) 10 (C) 14 (D) 20

7. What is the largest value of m such that the equation $\sin^{-1} x - mx = 0$ has three unique solutions? 1

(A) $m = 0$ (B) $m = 1$ (C) $m = \frac{\pi}{2}$ (D) $m = \pi$

8. $f(x)$ is shown in the following diagram, and $f(x)$ has a root at $x = x_0$. Sahil is trying to find the first approximation to this root, from where a second approximation will be found by Newton's Method. 1

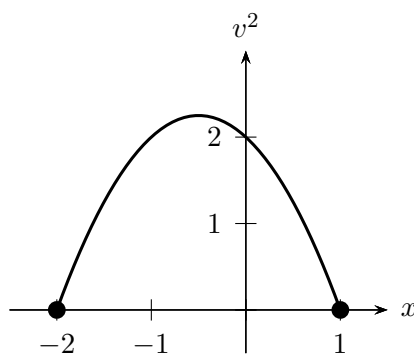


Which point should Sahil choose as his first approximation, so that his subsequent application of Newton's Method will produce a better approximation?

(A) A (B) B (C) C (D) D

9. A graph of the square of the velocity against displacement is shown.

1



What type of motion does this graph best describe?

- (A) Parabolic (C) Projectile
(B) Exponential (D) Simple harmonic
10. Which of the following is *not* equal to $\frac{d}{dx} \left(\frac{1}{2} v^2 \right)$? 1
- (A) $v \frac{dv}{dx}$ (B) $\frac{dv}{dt}$ (C) $\ddot{x}$ (D) $\dot{x}$

Examination continues overleaf...

Section II

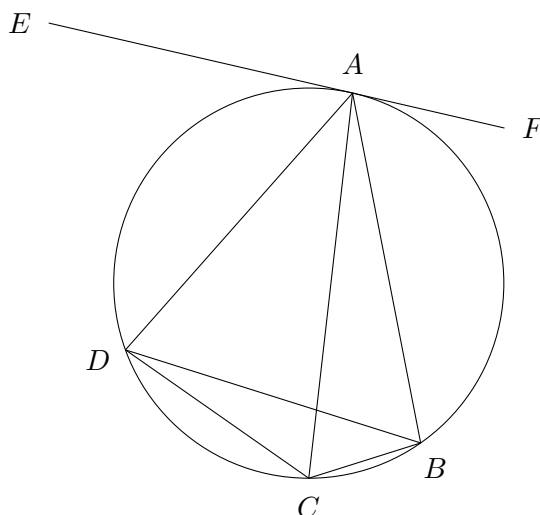
60 marks

Attempt Questions 11 to 14

Allow approximately 1 hour and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW booklet.	Marks
(a) Find the acute angle between the lines $2x - y = 0$ and $x - 2y = 0$, giving the answer correct to the nearest degree.		2
(b) Solve the inequality $\frac{x^2 - 4}{x} \geq 0$.		3
(c) Find the roots of the equation $x^3 + 6x^2 - x - 30 = 0$ given one of the roots of the equation is the sum of the other roots.		3
(d) $ABCD$ is a cyclic quadrilateral. EF is a tangent at A to the circle. CA bisects $\angle BCD$.		3

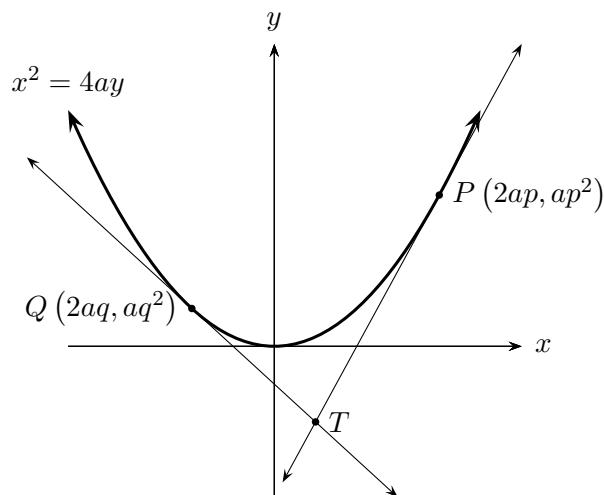


Show that $EF \parallel DB$.

Question 11 continued overleaf...

Question 11 continued from previous page...

- (e) The diagram shows the parabola $x^2 = 4ay$. The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola. The tangents at P and Q intersect at the point T .



- i. By referring to the Reference Sheet or otherwise, show that T has coordinates $(a(p+q), apq)$. 2
- ii. The points P and Q move on the parabola such that the gradient of the chord PQ is always equal to $\frac{a}{2}$. 2

Show that the locus of T is parallel to the axis of the parabola.

Examination continues overleaf...

Question 12 (15 Marks)

Commence a NEW booklet.

Marks

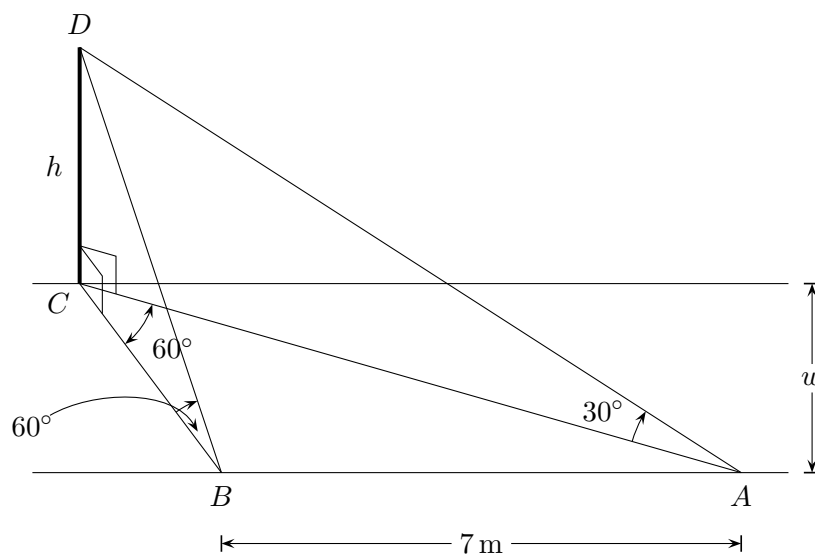
- (a) Prove by induction that:

3

$$\frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} + \cdots + \frac{2^n \times n}{(n+1)(n+2)} = \frac{2^{n+1}}{n+2} - 1$$

for all integers $n \geq 1$.

- (b) A footpath on horizontal ground has two parallel edges. CD is a vertical street sign pole of height h metres which stands with its base C on one edge of the footpath. A and B are two points on the other edge of the footpath such that $AB = 7$ m and $\angle ACB = 60^\circ$. From A and B , the angles of elevation to the top of the pole at D are 30° and 60° respectively.



- i. Show that the exact height of the flagpole is $h = \sqrt{21}$ metres. **3**
 - ii. By considering the area of $\triangle ABC$, find the exact width w of the footpath. **3**
- (c)
- i. Find the derivative of $f(x) = \tan(x^2)$. **2**
 - ii. Hence or otherwise, evaluate $\int_{-a}^a x \sec^2(x^2) dx$. **2**
- (d) Find the simplest expression for $\sin(\cos^{-1}(x-1))$ in terms of x only. **2**

Examination continues overleaf...

Question 13 (15 Marks) Commence a NEW booklet. **Marks**

(a) Evaluate $\int \frac{1}{\sqrt{9-4x^2}} dx$. **2**

(b) Use the substitution $u = 3 + e^x$ to find the exact value of **3**

$$\int_0^{\ln 5} \frac{e^x}{\sqrt{3+e^x}} dx$$

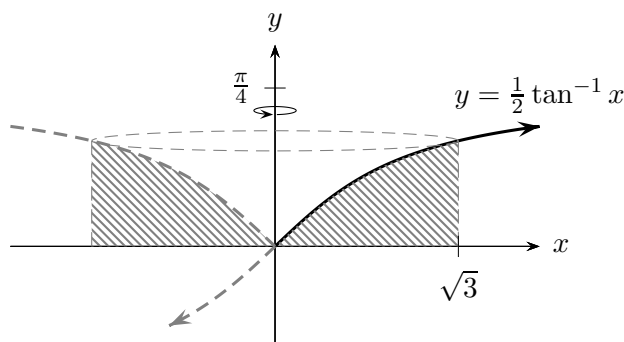
(c) i. Differentiate $y = \tan^{-1}(e^x)$. **2**

ii. Find the equation of the tangent to $y = \tan^{-1}(e^x)$ at $x = 0$. **2**

iii. Discuss the behaviour of the curve as $x \rightarrow \infty$. **1**

(d) i. Evaluate $\int \tan^2 2x dx$. **2**

ii. Find the exact volume generated when the area between curve $y = \frac{1}{2} \tan^{-1} x$, the x axis between $x = 0$ and $x = \sqrt{3}$ is rotated about the y axis. **3**



(Extension 2 students: Do NOT use volumes by slicing or cylindrical shells)

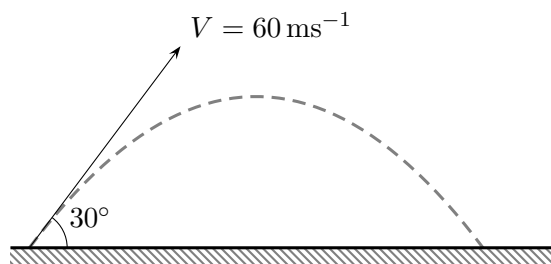
Examination continues overleaf...

Question 14 (15 Marks)

Commence a NEW booklet.

Marks

- (a) A projectile is launched from the origin, across a level plain at 30° to the horizontal and at an initial speed of 60 ms^{-1} . Use $g = 10 \text{ ms}^{-2}$.



The displacement equations of motion are:

$$\begin{cases} x = 30\sqrt{3}t \\ y = 30t - 5t^2 \end{cases} \quad (\text{Do NOT prove these})$$

- i. Find the maximum height of the particle. **2**
 - ii. Find the velocity of the particle one second after launch. **2**
- (b) A particle is moving in simple harmonic motion with

$$\ddot{x} = -4x + 4$$

Initially, the particle is at the origin and is moving away from the origin with a speed of $2\sqrt{3} \text{ ms}^{-1}$.

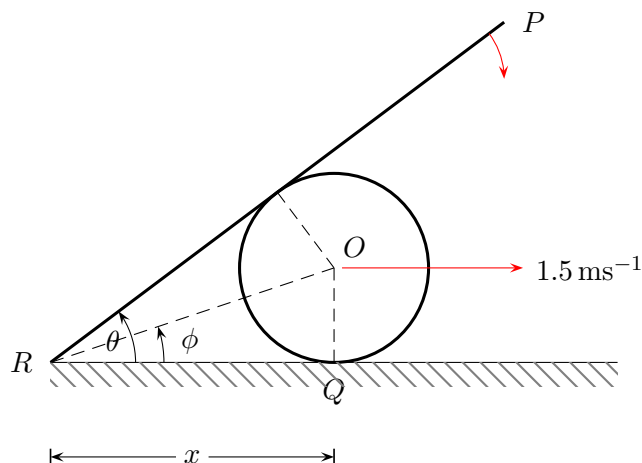
- i. Use integration to show that $v^2 = -4(x - 3)(x + 1)$. **2**
- ii. Find the centre and amplitude of the motion. **2**
- iii. Find the displacement-time equation for this particle. **3**

Question 14 continued overleaf...

Question 14 continued from previous page...

- (c) A rod RP is leaning on a disc of radius 1 m as shown in the diagram. The centre of the disc is labelled O .

One end of the rod R , is fixed to the ground, whilst the disc is tangential to the ground at Q .



The disc rolls to the right with a constant speed of 1.5 ms^{-1} . As the disc rolls, the other end of the rod P commences its descent towards the ground.

Let $\angle ORQ = \phi$ and $\angle PRQ = \theta$ and $RQ = x$.

- i. Write a relationship between $\angle ORQ$, OQ and RQ . 1
- ii. Hence or otherwise, find the rate of change of θ in radians per second, when $x = 6 \text{ m}$. 3

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

BOSTES NUMBER:

Class (please ✓)

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|-------------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 – | ☐ A | ☐ B | ☐ C | ☐ D |

2016 Mathematics Extension 1 HSC Course Trial Examination STUDENT SELF REFLECTION

1. In hindsight, did I do the best I can? Why or why not?

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2. Which topics did I need more help with, and what parts specifically?

- Q1-4, 8, 11 - Miscellaneous topics, Polynomials, Circle Geometry, Parametric Representation

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- Q5, 7, 12 - Induction, 3D Trigonometry, Inverse Trigonometry

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- Q6, 13 - Integration (of inverse trig/substitution/volumes), Curve Sketching.

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- Q9-10, 14 - Further applications of calculus to the physical world.

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3. What other parts from the feedback session can I take away to refine my solutions for future reference?

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Sample Band 6 Responses

Section I

1. (C) 2. (D) 3. (B) 4. (D) 5. (A)
6. (C) 7. (C) 8. (A) 9. (D) 10. (D)

Section II

Question 11 (Wall)

(a) (2 marks)

$$\begin{array}{l|l} 2x - y = 0 & x - 2y = 0 \\ y = 2x & y = \frac{1}{2}x \end{array}$$

$$\therefore m_1 = 2 \quad m_2 = \frac{1}{2}$$

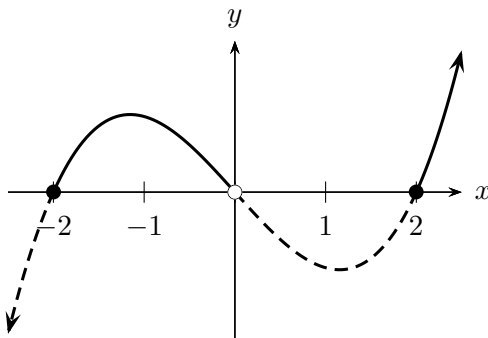
Apply angle between two lines formula,

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{2 - \frac{1}{2}}{1 + (2)\left(\frac{1}{2}\right)} \right| \\ &= \left| \frac{\frac{3}{2}}{2} \right| = \frac{3}{4} \\ \therefore \theta &= 36.869^\circ \dots \approx 37^\circ \end{aligned}$$

(b) (3 marks)

- ✓ [1] for multiplying by the square of the denominator.
- ✓ [1] for significant progress towards solution.
- ✓ [1] for removing $x = 0$ as a possible solution.

$$\begin{aligned} \frac{x^2 - 4}{x \times x^2} &\geq \frac{0}{x^2} \\ x(x^2 - 4) &\geq 0 \\ x(x - 2)(x + 2) &\geq 0 \end{aligned}$$



$$\therefore -2 \leq x < 0 \text{ or } x \geq 2$$

(c) (3 marks)

- ✓ [1] for correct usage of elementary symmetric functions.
- ✓ [1] for significant progress towards solution.
- ✓ [1] for correct final solutions.

$$x^3 + 6x^2 - x - 30 = 0$$

Roots are α , β and $\alpha + \beta$

- Sum of roots:

$$\begin{aligned} \alpha + \beta + (\alpha + \beta) &= -\frac{6}{1} = -6 \\ \therefore 2(\alpha + \beta) &= -6 \\ \alpha + \beta &= -3 \end{aligned}$$

- Product:

$$\begin{aligned} \alpha\beta(\alpha + \beta) &= -\frac{d}{a} = 30 \\ \alpha\beta(-3) &= 30 \\ \therefore \alpha\beta &= -10 \end{aligned}$$

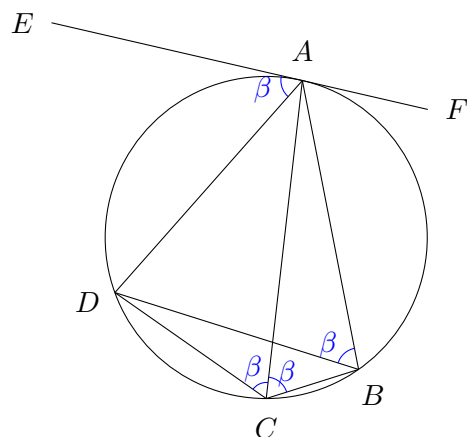
Form a quadratic with sum of roots -3 and product of roots -10 :

$$\begin{aligned} x^2 - (\alpha + \beta)x + \alpha\beta &= 0 \\ x^2 + 3x - 10 &= 0 \\ (x + 5)(x - 2) &= 0 \\ \therefore x &= 2, -5 \end{aligned}$$

Hence roots are $x = 2, -5, -3$.

(d) (3 marks)

- ✓ [1] for appropriate usage of $\angle$ in the alternate segment.
- ✓ [1] for appropriate usage of $\angle$ in the same segment.
- ✓ [1] for final justification.

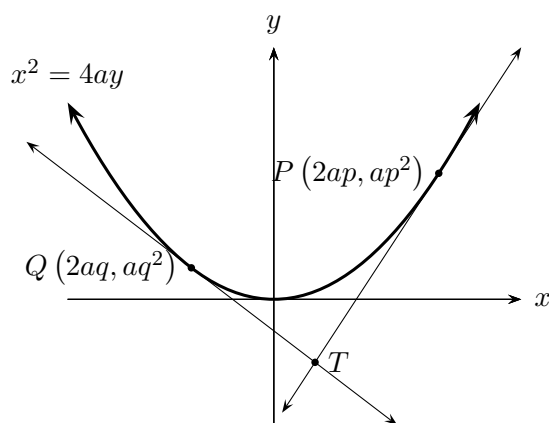


To prove $EF \parallel DB$: let $\angle BAF = \alpha$ and $\angle BCA = \angle ACD = \beta$.

- Let $\angle ACD = \beta$.
Then $\angle ACB = \beta$
(CA bisects $\angle BCD$, given)
- $\angle ABD = \angle ACD = \beta$
($\angle$ in the same segment, on chord AD).
- $\angle FAB = \angle ACB = \beta$
($\angle$ in the alternate segment)
- $\therefore EF \parallel DB$
(alternate angles $\angle FAB = \angle ABD$)

(e) i. (2 marks)

- ✓ [1] for adequate steps to show $x = a(p + q)$.
- ✓ [1] for adequate steps to show $y = apq$.



Solving simultaneously for equation

of tangent at P and Q

$$\begin{cases} y = px - ap^2 \\ y = qx - aq^2 \end{cases}$$

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p - q)(p + q)$$

$$\therefore x = a(p + q)$$

Substitute into $y = px - ap^2$:

$$\begin{aligned} y &= p(a)(p + q) - ap^2 \\ &= ap^2 + apq - ap^2 \\ &= apq \end{aligned}$$

ii. (2 marks)

- ✓ [1] for finding relationship between $p + q$ and a .
- ✓ [1] for correct justification of locus.

$$m_{PQ} = \frac{a}{2}$$

Applying gradient formula with $m = \frac{a}{2}$ between $P(2ap, ap^2)$ and $Q(2aq, aq^2)$:

$$\frac{aq^2 - ap^2}{2aq - 2ap} = \frac{a}{2}$$

$$\frac{a(q - p)(q + p)}{2a(q - p)} = \frac{a}{2}$$

$$\therefore p + q = a$$

Substituting into x_T :

$$x_T = a(p + q) = a^2$$

As the x coordinate at T is a constant, the locus of T is a straight line at $x = a^2$.

Question 12 (Lam)

- Examine $P(k+1)$:

(a) (3 marks)

- ✓ [1] for testing base case.
- ✓ [1] for correct use of the inductive hypothesis in the testing.
- ✓ [1] for final answer.

Let $P(n)$ be the proposition

$$\begin{aligned} \frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} \\ + \cdots + \frac{2^n \times n}{(n+1)(n+2)} \\ = \frac{2^{n+1}}{n+2} - 1 \end{aligned}$$

for all integers $n \geq 1$.

- Base case: $P(1)$

$$\begin{aligned} \text{Left: } \frac{2 \times 1}{2 \times 3} &= \frac{2}{6} \\ &= \frac{1}{3} \\ \text{Right: } \frac{2^{1+1}}{1+2} - 1 &= \frac{2^2}{3} - 1 \\ &= \frac{4}{3} - \frac{3}{3} \\ &= \frac{1}{3} \end{aligned}$$

Hence $P(1)$ is true.

- Inductive hypothesis: assume $P(k)$ is true for $k \in \mathbb{Z}^+$, i.e. $P(k)$ is

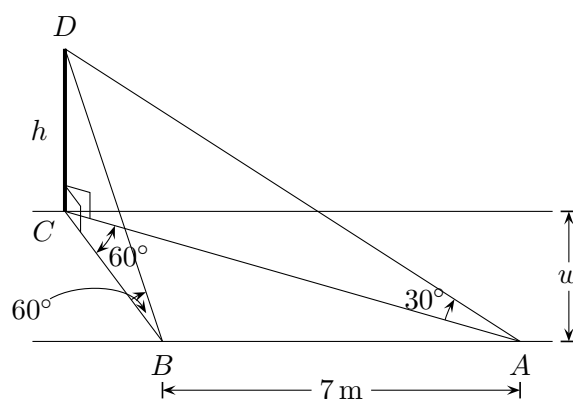
$$\begin{aligned} \frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} \\ + \cdots + \frac{2^k \times k}{(k+1)(k+2)} \\ = \frac{2^{k+1}}{k+2} - 1 \end{aligned}$$

$$\begin{aligned} \frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} \\ + \cdots + \frac{2^k \times k}{(k+1)(k+2)} \\ + \frac{2^{k+1}(k+1)}{(k+2)(k+3)} \\ = \frac{2^{k+1} \times (k+3)}{k+2} - 1 + \frac{2^{k+1}(k+1)}{(k+2)(k+3)} \\ = \frac{2^{k+1}(k+3) + 2^{k+1}(k+1)}{(k+2)(k+3)} - 1 \\ = \frac{2^{k+1}(2k+4)}{(k+2)(k+3)} - 1 \\ = \frac{2 \times 2^{k+1} \cancel{(k+2)}}{\cancel{(k+2)}(k+3)} - 1 \\ = \frac{2^{(k+1)+1}}{(k+1)+2} - 1 \end{aligned}$$

 $\therefore P(k+1)$ is also true, and $P(n)$ is true by induction.

(b) i. (3 marks)

- ✓ [1] for both CB and CA in terms of h .
- ✓ [1] for correct usage of cosine rule in $\triangle ABC$.
- ✓ [1] for demonstrating $h = \sqrt{21}$.



- In $\triangle DCB$,

$$\begin{aligned} \frac{h}{CB} &= \tan 60^\circ = \sqrt{3} \\ \therefore CB &= \frac{h}{\sqrt{3}} \end{aligned}$$

- In $\triangle DCA$,

$$\frac{h}{CA} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\therefore CA = h\sqrt{3}$$

- Applying cosine rule in $\triangle ABC$:

$$AB^2 = BC^2 + AC^2 - 2(BC)(AC) \cos 60^\circ \text{ii. (2 marks)}$$

$$7^2 = \frac{h^2}{3} + 3h^2 - 2h^2 \times \frac{1}{2}$$

$$\therefore 49 = \frac{7}{3}h^2$$

$$h^2 = 21$$

$$h = \sqrt{21}$$

- ii. (3 marks)

- ✓ [1] for area by sine rule
- ✓ [1] for area by $\frac{1}{2} \times h_{\perp} \times \text{base}$
- ✓ [1] for final value of w .
- Finding the area by base and perpendicular height, where the perpendicular height is denoted w :

$$A = \frac{1}{2} \times 7 \times w$$

- Finding the area by the sine rule:

$$CB = \frac{\sqrt{21}}{\sqrt{3}} = \sqrt{7}$$

$$CA = \sqrt{21}\sqrt{3} = 3\sqrt{7}$$

$$A = \frac{1}{2}ab \sin C$$

$$= \frac{1}{2}(BC)(AC) \sin 60^\circ$$

$$= \frac{1}{2}(\sqrt{7})(3\sqrt{7}) \frac{\sqrt{3}}{2}$$

Equating areas,

$$\frac{7}{2}w = \frac{1}{2} \times (\sqrt{7})(3\sqrt{7}) \frac{\sqrt{3}}{2}$$

$$\frac{7}{2}w = \frac{7}{4} \times 3\sqrt{3}$$

$$w = \frac{3\sqrt{3}}{2} \text{ m}$$

- (c) i. (2 marks)

$$f(x) = \tan(x^2)$$

Applying chain rule,

$$f'(x) = \sec^2(x^2) \times 2x$$

$$= 2x \sec^2(x^2)$$

$$\int_{-a}^a x \sec^2(x^2) dx = 0$$

($y = x$ is odd, $y = \sec^2(x^2)$ is even, i.e. odd $\times$ even = odd. Integrating an odd function over a balanced interval $-a < x < a$ produces zero.)

Alternatively,

$$\int_{-a}^a x \sec^2(x^2) dx = \frac{1}{2} [\tan(x^2)]_{-a}^a$$

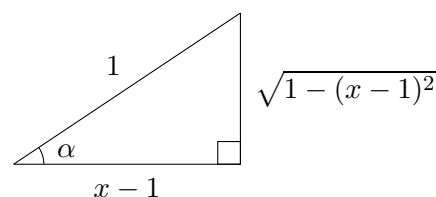
$$= \frac{1}{2} (\tan a^2 - \tan a^2)$$

$$= 0$$

- (d) (2 marks)

- ✓ [1] for relationship between $\cos \alpha$ and x .
- ✓ [1] for simplest expression.

Let $\alpha = \cos^{-1}(x-1)$. Hence $\cos \alpha = x-1$. Draw a right $\angle$ triangle with adjacent side to α of length $x-1$ and unit hypotenuse.



$$\therefore \sin(\cos^{-1}(x-1)) = \sin \alpha$$

$$= \sqrt{1 - (x-1)^2}$$

$$= \sqrt{(1 - (x-1))(1 + (x-1))}$$

$$= \sqrt{(2-x)x}$$

Question 13 (Lawson)

(a) (2 marks)

$$\begin{aligned}\int \frac{1}{\sqrt{9-4x^2}} dx &= \int \frac{1}{\sqrt{4\left(\frac{9}{4}-x^2\right)}} dx \\ &= \frac{1}{2} \sin^{-1}\left(\frac{2x}{3}\right) + C\end{aligned}$$

(b) (3 marks)

- ✓ [1] for correct relationship between differentials.
- ✓ [1] for correct primitive.
- ✓ [1] for final answer.

$$\int_0^{\ln 5} \frac{e^x}{\sqrt{3+e^x}} dx$$

With substitution $u = 3 + e^x$:

$$\begin{aligned}\frac{du}{dx} &= e^x \\ \therefore du &= e^x dx \\ x=0 \quad u &= 3 + e^0 = 4 \\ x=\ln 5 \quad u &= 3 + e^{\ln 5} = 8 \\ \int_0^{\ln 5} \frac{e^x}{\sqrt{3+e^x}} dx &= \int_{u=4}^{u=8} \frac{du}{\sqrt{u}} \\ &= \int_4^8 u^{-\frac{1}{2}} du \\ &= \left[2u^{\frac{1}{2}}\right]_4^8 \\ &= 2(\sqrt{8} - \sqrt{4}) \\ &= 2(2\sqrt{2} - 2) \\ &= 4(\sqrt{2} - 1)\end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned}y &= \tan^{-1}(e^x) \\ \frac{dy}{dx} &= \frac{1}{1+(e^x)^2} \times e^x = \frac{e^x}{1+e^{2x}}\end{aligned}$$

ii. (2 marks)

- ✓ [-1] for each newly introduced error.

At $x = 0$

$$\begin{aligned}y &= \tan^{-1} e^0 = \tan^{-1} 1 \\ &= \frac{\pi}{4}\end{aligned}$$

Finding the gradient at $x = 0$:

$$\frac{dy}{dx} = \frac{e^0}{1+e^0} = \frac{1}{2}$$

Applying point-gradient formula,

$$\begin{aligned}\frac{y - \frac{\pi}{4}}{x - 0} &= \frac{1}{2} \\ y - \frac{\pi}{4} &= \frac{1}{2}x \\ \therefore y &= \frac{1}{2}x + \frac{\pi}{4}\end{aligned}$$

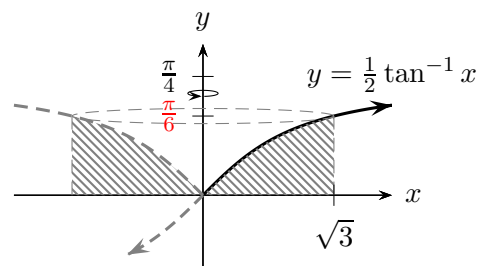
iii. (1 mark) As $x \rightarrow \infty$, $e^x \rightarrow \infty$.
Hence $y = \tan^{-1}(e^x) \rightarrow \frac{\pi}{2}$.

(d) i. (2 marks)

$$\begin{aligned}\int \tan^2 2x dx &= \int (\sec^2 2x - 1) dx \\ &= \frac{1}{2} \tan 2x - x + C\end{aligned}$$

ii. (3 marks)

- ✓ [1] for changing subject to x , and limits of integration along y axis.
- ✓ [1] for correct primitive
- ✓ [1] for final answer.



When $x = \sqrt{3}$, $y = \frac{1}{2} \tan^{-1} \sqrt{3} = \frac{\pi}{6}$.
Changing subject from x to y :

$$\begin{aligned}y &= \frac{1}{2} \tan^{-1} x \\ 2y &= \tan^{-1} x \\ \therefore x &= \tan 2y\end{aligned}$$

Volume by rotating along y axis:

$$\begin{aligned} V_1 &= \pi \int_a^b x^2 dy \\ &= \pi \int_0^{\frac{\pi}{6}} \tan^2 2y dy \\ &= \pi \left[\frac{1}{2} \tan 2y - y \right]_0^{\frac{\pi}{6}} \\ &= \pi \left(\frac{1}{2} \tan \frac{\pi}{3} - \frac{\pi}{6} \right) \\ &= \pi \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) \end{aligned}$$

Finding the volume of the cylinder,

$$\begin{aligned} V_{\text{cylinder}} &= \pi r^2 h \\ &= \pi \left(\sqrt{3} \right)^2 \times \frac{\pi}{6} \\ &= \frac{\pi^2}{2} \end{aligned}$$

Volume generated (shaded):

$$\begin{aligned} V &= V_{\text{cylinder}} - V_1 \\ &= \frac{\pi^2}{2} - \left(\pi \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) \right) \\ &= \frac{4\pi^2}{6} - \frac{\pi\sqrt{3}}{2} \\ &= \frac{2\pi^2}{3} - \frac{\pi\sqrt{3}}{2} \end{aligned}$$

Question 14 (Gan)

(a) i. (2 marks)

✓ [1] for identifying parameters for maximum height.

✓ [1] for finding value of maximum height.

Maximum height occurs when $\dot{y} = 0$:

$$\begin{aligned} y &= 30t - 5t^2 \\ \dot{y} &= 30 - 10t = 0 \\ \therefore 10t &= 30 \\ t &= 3 \end{aligned}$$

Substitute into y :

$$\begin{aligned} y &= 30(3) - 5(3^2) \\ &= 90 - 45 \\ &= 45 \text{ m} \end{aligned}$$

ii. (2 marks)

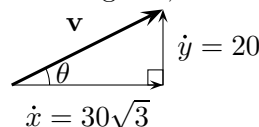
✓ [1] for correct vector diagram

✓ [1] for magnitude and direction of velocity.

When $t = 1$:

$$\begin{aligned} \dot{x} &= 30\sqrt{3} \\ \dot{y} &= 30 - 10t \Big|_{t=1} \\ &= 20 \end{aligned}$$

Drawing vector diagram,



$$\begin{aligned} v^2 &= (20)^2 + (30\sqrt{3})^2 \\ &= 400 + 2700 \\ &= 3100 \\ &= 10\sqrt{31} \text{ ms}^{-1} \end{aligned}$$

Finding the angle at $t = 1$

$$\begin{aligned} \tan \theta &= \frac{20}{30\sqrt{3}} \\ \theta &= 21^\circ 3' \end{aligned}$$

(b) i. (2 marks)

✓ [1] for integrating to obtain $\frac{1}{2}v^2$

✓ [1] for showing required result.

$$\ddot{x} = -4x + 4$$

Integrating,

$$\begin{aligned} \frac{1}{2}v^2 &= \int (-4x + 4) dx \\ &= -2x^2 + 4x + C_1 \\ v^2 &= -4x^2 + 8x + C_2 \end{aligned}$$

When $t = 0$, $v^2 = 4 \times 3 = 12$, $x = 0$:

$$\begin{aligned} 12 &= -4(0) + 8(0) + C_2 \\ \therefore C_2 &= 12 \\ \therefore v^2 &= -4x^2 + 8x + 12 \\ &= -4(x^2 - 2x - 3) \\ &= -4(x - 3)(x + 1) \end{aligned}$$

ii. (2 marks)

- Extremities of motion at $x = 3$ and $x = -1$.
- Centre of motion at $x = \frac{3-1}{2} = 1$
- Amplitude $a = 2$

iii. (3 marks)

- ✓ [1] for obtaining completed square form
- ✓ [1] for obtaining an inverse trigonometric primitive
- ✓ [1] for correct equation

Completing the square,

$$\begin{aligned} v^2 &= -4(x^2 - 2x + 1 - 4) \\ &= -4((x-1)^2 - 4) \\ &= 4(4 - (x-1)^2) \\ \therefore v &= \pm 2\sqrt{4 - (x-1)^2} \end{aligned}$$

As v is non determinant due to initial conditions, assume positive root and perform separation of variables:

$$\begin{aligned} \frac{dx}{dt} &= 2\sqrt{4 - (x-1)^2} \\ \frac{dx}{\sqrt{4 - (x-1)^2}} &= 2 dt \end{aligned}$$

Integrating,

$$\begin{aligned} \int \frac{dx}{\sqrt{4 - (x-1)^2}} &= \int 2 dt \\ \sin^{-1}\left(\frac{x-1}{2}\right) &= 2t + C_3 \end{aligned}$$

When $t = 0$, $x = 0$:

$$\begin{aligned} \sin^{-1}\left(-\frac{1}{2}\right) &= C_3 \\ \therefore C_3 &= -\frac{\pi}{6} \\ \frac{x-1}{2} &= \sin\left(2t - \frac{\pi}{6}\right) \\ x &= 1 + 2\sin\left(2t - \frac{\pi}{6}\right) \end{aligned}$$

Alternatively, given the initial velocity was not defined in terms of direction, students may have obtained a different phase shift when

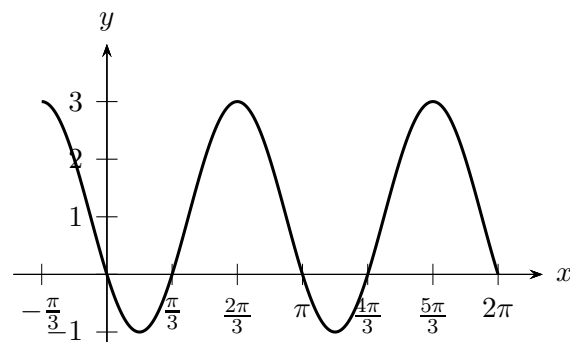
performing the steps above and assumed a negative root. This results in $x = 2\cos\left(2t - \frac{2\pi}{3}\right) + 1$.

Alternatively, sketch a graph assuming a cosine as the sinusoid with centre of motion at $x = 1$, and use initial conditions $t = 0$, $x = 0$ to find the phase shift:

$$x = 2\cos(2t + \alpha) + 1$$

When $t = 0$, $x = 0$:

$$\begin{aligned} 0 &= 2\cos\alpha + 1 \\ \cos\alpha &= -\frac{1}{2} \\ \alpha &= \frac{2\pi}{3} \\ x &= 2\cos\left(2t + \frac{2\pi}{3}\right) + 1 \end{aligned}$$

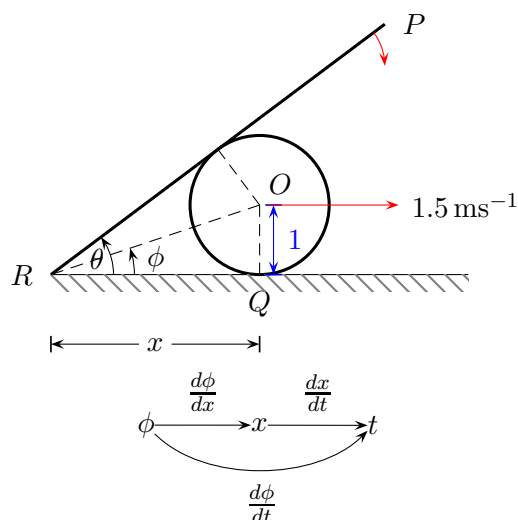


(c) i. (1 mark)

- $OQ = 1$
- $\tan\phi = \frac{1}{x}$

ii. (3 marks)

- ✓ [1] correct derivative $\frac{d\phi}{dx}$.
- ✓ [1] for finding $\frac{d\phi}{dx} = -\frac{1}{37}$.
- ✓ [1] for finding $\frac{d\theta}{dt}$.



Applying chain rule,

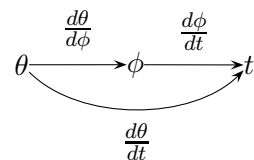
$$\frac{d\phi}{dt} = \frac{d\phi}{dx} \times \frac{dx}{dt}$$

Given $\tan \phi = \frac{1}{x}$, rearranging to find $\frac{d\phi}{dx}$:

$$\begin{aligned}\phi &= \tan^{-1}(x^{-1}) \\ \frac{d\phi}{dx} &= \frac{1}{1 + (x^{-1})^2} \times -x^{-2} \\ &= \frac{1}{1 + \frac{1}{x^2}} \times \frac{-1}{x^2} = -\frac{1}{x^2 + 1} \Big|_{x=6} \\ &= -\frac{1}{37}\end{aligned}$$

Evaluating $\frac{d\phi}{dt}$:

$$\frac{d\phi}{dt} = -\frac{1}{37} \times \frac{3}{2} = -\frac{3}{2 \times 37}$$



Using the $\theta - \phi$ chain,

$$\theta = 2\phi$$

$$\therefore \frac{d\theta}{d\phi} = 2$$

$$\begin{aligned}\frac{d\theta}{dt} &= \frac{d\theta}{d\phi} \times \frac{d\phi}{dt} \\ &= 2 \times -\frac{3}{2 \times 37} \\ &= -\frac{3}{37} \text{ rad/s}\end{aligned}$$